

A Data and Candidate Registry

Endpoint registration



9 endpoints
curated · no overlap

Fixed candidate registry



32 candidates · fixed order

B Repeated Nested Scaffold Selection

Outer/inner partitions



10 split seeds · 3 outer scaffold folds · selection confined to training data

Fold-specific preprocessing



Inner-validation ranking

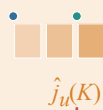


Freeze decision + outer audit



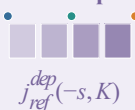
C Frozen References and Completion-Gap Decomposition

Validation-selected candidate



$\hat{j}_u(K)$

K-dependent within-prefix reference



$j_{ref}^{dep}(-s, K)$

K-invariant full-registry reference



$j_{ref}^{inv}(-s)$

$$\Delta_{inv} = \frac{1}{S} \sum_S [G_{inv}(s, 32) - G_{inv}(s, 4)]$$
$$G_{inv} = G_{avail} + G_{dep}$$

D Robustness and Chemical Support

Ranking & equivalence



CAHit@3 · τ

Utility-pattern diversity



rank · entropy

Transfer & composition



split · equal size

Chemical support & error audit



novel · error · FN

E Auditable Evidence and Reporting

- Inner and outer probabilities
- Eligibility and frozen identities
- Splits, settings and hashes
- Failures and uncertainty
- Machine-readable release

Auditable decision record

Registry

Nested selection

Gap decomposition

Robustness

Reporting